EXP NO: 1 Date:16-7-26	SETTING UP THE ENVIRONMENT AND PREPROCESSING THE DATA

AIM:

To set up a fully functional machine learning development environment and to perform data preprocessing operations like handling missing values, encoding categorical variables, feature scaling, and splitting datasets.

ALGORITHM:

1. Install Required Libraries:

• Install numpy, pandas, matplotlib, seaborn, and scikit-learn using pip.

2. Import Libraries.

3. Load Dataset:

• Load any dataset (e.g., Titanic or Iris) using pandas.

4. Data Exploration:

• Use df.info(), df.describe(), df.isnull().sum() to understand the data.

5. Handle Missing Values:

• Use .fillna() or .dropna() depending on the strategy.

6. Encode Categorical Data:

• Use pd.get_dummies() or LabelEncoder.

7. Feature Scaling:

• Normalize or standardize the numerical features using StandardScaler or

MinMaxScaler.

8. Split Dataset:

• Use train_test_split() from sklearn to create training and testing sets.

9. Display the Preprocessed Data.

CODE:

preprocessing Data.py - C:/Users/HDCO422208/Downloads/preprocessing Data.py (3.14.4)

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```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler, LabelEncoder
import matplotlib.pyplot as plt

df = pd.read_csv(r"C:\Users\HDCO422208\Downloads\titanic.csv")

print("df.head()")
print(df.head())
print("\n" + "="*50 + "\n")

print("df.info()")
df.info()
print("\ndf.describe()")
print(df.describe())
print("\ndf.isnull().sum()")
print(df.isnull().sum())
print("\n" + "="*50 + "\n")

df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

df.drop(columns=['Cabin'], inplace=True)

le = LabelEncoder()
df['Sex'] = le.fit_transform(df['Sex'])
df['Embarked'] = le.fit_transform(df['Embarked'])

df.drop(columns=['PassengerId', 'Name', 'Ticket'], inplace=True)

scaler = StandardScaler()
numerical_cols = ['Age', 'Fare']
df[numerical_cols] = scaler.fit_transform(df[numerical_cols])

X = df.drop('Survived', axis=1)
y = df['Survived']
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

print("Final Processing Shapes")
print("Training Data Shape:", X_train.shape)
print("Test Data Shape:", X_test.shape)
print("\nPreprocessed X_train: this ois the foirst five rows training")
print(X_train.head())
```

OUTPUT:

```
========================================================================================= RESTART: C:/Users/HDCO4
df.head()
   PassengerId  Survived  Pclass  ...     Fare Cabin  Embarked
0            1         0       3  ...   7.2500   NaN         S
1            2         1       1  ...  71.2833   C85         C
2            3         1       3  ...   7.9250   NaN         S
3            4         1       1  ...  53.1000  C123         S
4            5         0       3  ...   8.0500   NaN         S

[5 rows x 12 columns]

==================================================

df.info()
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB

df.describe()
       PassengerId    Survived      Pclass  ...       SibSp       Parch        Fare
count   891.000000  891.000000  891.000000  ...  891.000000  891.000000  891.000000
mean    446.000000    0.383838    2.308642  ...    0.523008    0.381594   32.204208
std     257.353842    0.486592    0.836071  ...    1.102743    0.806057   49.693429
min       1.000000    0.000000    1.000000  ...    0.000000    0.000000    0.000000
25%     223.500000    0.000000    2.000000  ...    0.000000    0.000000    7.910400
50%     446.000000    0.000000    3.000000  ...    0.000000    0.000000   14.454200
75%     668.500000    1.000000    3.000000  ...    1.000000    0.000000   31.000000
max     891.000000    1.000000    3.000000  ...    8.000000    6.000000  512.329200

[8 rows x 7 columns]

df.isnull().sum()
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
```

```
=========================================================================
Final Processing Shapes
Training Data Shape: (712, 7)
Test Data Shape: (179, 7)

Preprocessed X_train: this ois the foirst five rows training
     Pclass  Sex       Age  SibSp  Parch      Fare  Embarked
331       1    1  1.240235      0      0 -0.074583         2
733       2    1 -0.488887      0      0 -0.386671         2
382       3    1  0.202762      0      0 -0.488854         2
704       3    1 -0.258337      1      0 -0.490280         2
813       3    0 -1.795334      4      2 -0.018709         2
>>
```

RESULT:

The Python environment was successfully set up and the dataset was pre-processed by handling missing values, encoding categorical data, performing feature scaling, and splitting the data into training and testing sets. The dataset is now ready for model training and analysis.